

Online Resource 2 - detailed tables and extended evidence

This supplementary document retains the detailed tables, extended evidence hierarchy, model-component documentation and secondary validation material moved out of the main manuscript to respect the International Journal of Geotechnical Engineering research-article word limit. The main manuscript remains the authoritative narrative; this file preserves auditability and table traceability.

Detailed Tables from Full Technical Version

Table 1a. Fit-to-journal relevance map for the International Journal of Geotechnical Engineering.

Journal interest	Manuscript element
Laterally loaded pile groups	ERIES-aligned fixed 2x2 pile-group screening case
Soil-pile interaction	p-y screening, group-interaction factor and continuum/interface checks
Cyclic loading/degradation	degradation-aware serviceability and feasibility evaluation
Numerical modelling	CalculiX, OpenSeesPy and localized 3D interface-layer evidence
Innovative applications	claim-level V4-S passport and Claim Support Index for pre-design decisions
Design practice	D-L-s/D candidate selection, active constraint diagnosis and escalation rules
Reproducibility	CSV files, scripts, figures, QA gates and supplementary screening capsule

Table 2a. Dimensional consistency audit for the equivalent-stiffness screening branch.

Symbol	Expected unit	Expression	Dimensional result	Consistent?	Correction
EI	force x length ²	pile bending stiffness	F L ²	Yes	none
k	force/length ³	subgrade modulus	F L ⁻³	Yes	state convention
T	length	$(EI/(kD))^{0.25}$	L	Yes	none
Le	length	min(L, 3.5T)	L	Yes	screening cutoff
k _{single}	force/length	$3EI/Le^3$	F L ⁻¹	Yes	none
K _{group}	force/length	n eta _g k _{single}	F L ⁻¹	Yes	eta _g dimensionless
u	length	H/K _{group}	L	Yes	screening displacement
C _{norm}	index	C _{raw} /1000	normalized index	Yes	not price
g _j	dimensionless	demand/capacity	1	Yes	keep <= 1

Table 1. Constraint definitions used in the degradation-aware optimization framework.

Constraint	Screening expression	Limit	Purpose
Pile-cap displacement	uH / ulim	1.0	Controls lateral serviceability
Pile-cap rotation	theta / thetalim	1.0	Controls serviceability and superstructure compatibility
Bending stress	sigma / sigmalim	1.0	Prevents structurally unrealistic pile demand
Soil reaction	Hpile / pultDLe	1.0	Checks mobilized soil reaction against degraded resistance

Constraint	Screening expression	Limit	Purpose
Fatigue	(stress ratio / reference) ³ N/Nref	1.0	Flags cyclic structural accumulation

Table 2 defines the design variables and bounds evaluated in the reproducible fixed-topology search. The present optimization is intentionally restricted to the ERIES-aligned 2x2 topology. Diameter, length and spacing are optimized, but topology is not varied in the final design space. Table 3 defines the idealized three-layer soil profile. The values are not presented as a site investigation; they define a consistent computational example that can be rerun and modified. Supplementary material includes the Python code, CSV outputs, figure files, a limited Rollins liquefied-sand p-y scale check that is not profile matched, a holdout p-y calibration capsule and a limited ERIES-RESPOND 2x2 acceleration-based dynamic-impedance consistency check and GEOLAB group-response calibration needed to reproduce the checks.

dimension	values	count
Topology	2x2	1
Diameter	0.60, 0.75, 0.90, 1.05, 1.20 m	5
Length	8, 10, 12, 14, 16, 18, 20 m	7
Spacing	2.5, 3.0, 3.5, 4.0, 4.5 D	5
Total	1 x 5 x 7 x 5	175

Table 2b. Cardinality of the fixed 2x2 design enumeration.

result	text	table	figure	csv_script	matches	action
fixed layout	2x2 fixed	Table 5	Fig. 2	canonical_2x2_design_space.csv	yes	retained
D	0.60 m	Table 5	Fig. 2	selected_candidate.csv	yes	verified
L	8.0 m	Table 5	Fig. 2	selected_candidate.csv	yes	verified
spacing	2.5D	Table 5	Fig. 2	selected_candidate.csv	yes	verified
u_CalculiX	3.219 mm	Table 20/21	Fig. 18/19	calcix_checks_trace.csv	yes	verified
u_FEM anchored	4.83 mm	Table 21b	Fig. 19	gamma_fem_sensitivity.csv	yes	verified
anchor band	3.219-6.439 mm	Table 21c	Fig. 19	gamma_fem_sensitivity.csv	yes	verified
u_stress	25.81 mm	Table 5/21c	Fig. 19	canonical_2x2_design_space.csv	yes	verified
utilization	0.860	Table 5	Fig. 5	canonical_2x2_design_space.csv	yes	verified
C_raw	12116.2	cost trace	Fig. 3	canonical_2x2_design_space.csv	yes	verified
C_norm	12.1	Table 5	Fig. 3	canonical_2x2_design_space.csv	yes	verified
alternatives	175	Table 2	none	01_generate_design_space.py	yes	verified
feasible alternatives	14	canonical CSV	Fig. 3	02_evaluate_candidates.py	yes	verified
reserve change	changes at 0.85	Table 7	Fig. 6	05_reserve_check.py	yes	verified
cycle multiplier	spacing changes at 4.0x	Table 8	Fig. 7	06_degradation_sweep.py	yes	verified
cost weights	candidate unchanged	Table 9	none	07_cost_weight_sensitivity.py	yes	verified

Table 2c. Reproducibility and traceability checks for the canonical 2x2 results.

Variable	Values	Role
Topology	2x2 only	Fixed ERIES-aligned layout; topology is not varied in the final reproduced search
Pile diameter D	0.60, 0.75, 0.90, 1.05, 1.20 m	Size variable within the fixed topology

Variable	Values	Role
Pile length L	8, 10, 12, 14, 16, 18, 20 m	Embedded-length variable
Spacing	2.5D, 3.0D, 3.5D, 4.0D, 4.5D	Interaction and cap-footprint variable
Load case	H = 100 kN; M = 0 kN m; N = 100,000 cycles	Cyclic lateral service demand

Table 2. Design variables used in the fixed 2x2 ERIES-aligned diameter-length-spacing search.

Layer	Depth (m)	Unit weight (kN/m ³)	Static k (MN/m ³)	pult (kPa)	alpha
Loose silty sand	0-4	18.0	18	145	0.18
Medium dense sand	4-11	19.0	45	320	0.14
Stiff sandy clay	11-22	19.5	70	520	0.10

Table 3. Idealized soil profile used for the degradation-aware p-y screening calculation.

Metric	Value
Topology	2x2
Number of piles	4
Diameter	0.60 m
Length	8.0 m
Spacing	2.5D (1.50 m)
Group efficiency	0.694
FEM-anchored preliminary displacement	4.83 mm
Deterministic screening band	3.219-6.439 mm
Severe-degradation stress-test displacement	25.81 mm
Governing utilization	0.860, based on the 25.81 mm stress-test envelope
Preliminary-prediction displacement utilization	0.161, based on 4.83 mm / 30 mm
Raw weighted cost C _{raw}	12116.2
Normalized cost index C _{norm} = C _{raw} / 1000	12.1

Table 4. Selected least-cost feasible screening candidate, governing utilization and normalized cost index.

Topology	Status	D (m)	L (m)	Spacing	FEM-anchored disp. (mm)	Stress-test disp. (mm)	Utilization basis	Normalized cost index C _{norm}
2x2	Feasible	0.60	8	2.5D	4.83	25.81	0.860 from stress-test envelope	12.1

Table 5. Selected least-cost feasible candidate within the 2x2 ERIES-aligned design space.

Case	Topology	D (m)	L (m)	Spacing	Soil k (kN/m ³)	T (m)	Le (m)	Utilization	Normalized cost index C _{norm}
no cyclic degradation	2x2	0.60	8	2.5D	23220	1.92	6.7	0.640	12.1
design cyclic degradation	2x2	0.60	8	2.5D	15649	2.12	7.4	0.860	12.1

Table 6. Baseline comparison with and without cyclic degradation. Cyclic degradation increases utilization from 0.640 to 0.860 while keeping the fixed 2x2 layout and normalized cost index C_{norm} unchanged.

Reserve limit	Topology	D (m)	L (m)	Spacing	Utilization	Normalized cost index C_{norm}
1.00	2x2	0.60	8	2.5D	0.860	12.1
0.95	2x2	0.60	8	2.5D	0.860	12.1
0.90	2x2	0.60	8	2.5D	0.860	12.1
0.85	2x2	0.60	8	3.0D	0.781	12.2

Table 7 and Fig. 6 address the concern that a near-limit design may be numerically fragile. For utilization limits of 1.00, 0.95 and 0.90, the selected ERIES-aligned 2x2 candidate remains unchanged with normalized cost index $C_{norm} = 12.1$ and utilization 0.860. Only when the reserve limit is tightened to 0.85 does the selected spacing change from 2.5D to 3.0D, increasing C_{norm} slightly to 12.2.

Cycle multiplier	Fixed layout	D (m)	L (m)	Spacing	Soil k (kN/m3)	Stress-test disp. (mm)	Feasible designs	Utilization	Normalized cost index C_{norm}
0.00	2x2	0.60	8	2.5D	23220	19.20	175	0.640	12.1
0.25	2x2	0.60	8	2.5D	18889	22.42	175	0.747	12.1
0.50	2x2	0.60	8	2.5D	17287	23.96	175	0.799	12.1
1.00	2x2	0.60	8	2.5D	15649	25.81	175	0.860	12.1
2.00	2x2	0.60	8	2.5D	14075	27.95	175	0.932	12.1
4.00	2x2	0.60	8	3.0D	12614	27.54	174	0.918	12.2

Table 7. Robust-reserve check for stricter utilization limits within the ERIES-aligned 2x2 design space.

Cost-weight case	Fixed layout	D (m)	L (m)	s/D	Rewighted normalized index $C_{norm,w}$	Comparable to baseline	Utilization
baseline weights	2x2	0.60	8.0	2.5	12.1	yes	0.860
concrete weight -20%	2x2	0.60	8.0	2.5	10.7	no; weights changed	0.860
concrete weight +20%	2x2	0.60	8.0	2.5	13.5	no; weights changed	0.860
installation weight -20%	2x2	0.60	8.0	2.5	11.2	no; weights changed	0.860
installation weight +20%	2x2	0.60	8.0	2.5	13.0	no; weights changed	0.860
cap-footprint weight -20%	2x2	0.60	8.0	2.5	12.0	no; weights changed	0.860
cap-footprint weight +20%	2x2	0.60	8.0	2.5	12.2	no; weights changed	0.860

Table 8. Sensitivity of selected screening candidate and feasible-count to cyclic-degradation multiplier.

Reference case	Topology	Spacing	Reported/anchor	Screening	Abs. diff.
ERIES-RESPOND full-scale 2x2 geometry, 2.5D	2x2	2.50D	-	0.694	-
Rollins et al. 2005 spacing-effect trend, 3.3D	2x2	3.30D	-	0.799	-
monotonic spacing trend	2x2	3.00D	-	0.765	-
monotonic spacing trend	2x2	3.30D	-	0.799	-
monotonic spacing trend	2x2	4.00D	-	0.861	-
monotonic spacing trend	2x2	5.00D	-	0.918	-
monotonic spacing trend	2x2	5.65D	-	0.942	-
monotonic spacing trend	2x2	6.00D	-	0.952	-

Table 9. Cost-weight sensitivity for the normalized objective function. Reweighted values are not directly comparable to the baseline except for the baseline-weight row.

Topology	Stress-test disp. (mm)	Independent Winkler disp. (mm)	Winkler/stress-test ratio	Role
2x2	25.81	2.55	0.10	single-case magnitude and stiffness cross-check

Table 10. Quantitative external trend-consistency check for spacing-dependent group efficiency.

Method	Source/provenance	Group interaction	Cyclic degradation	p-multiplier or eta_g	Head stiffness / solver	Predicted head disp. (mm)	Ratio to 4.83 mm	Ratio to 25.81 mm	Ratio to CalculiX p-y	Maximum defensible claim
Single-pile p-y baseline without group reduction	same degraded profile and stiffness expression; eta_g set to 1.0	none	0.674	1.000	5580.0	17.921	3.710	0.694	5.567	single-pile/no-group-reduction reference for model-to-model comparison only
Proposed screening stress-test envelope	this paper, eta_g equation and degraded profile	eta/p-multiplier	0.674	0.694	3873.9	25.814	5.344	1.000	8.018	conservative screening envelope for selected 2x2 benchmark
Proposed FEM-anchored preliminary estimate	1.5 times CalculiX p-y FEM result	eta/p-multiplier	not implemented	-	20703.9	4.830	1.000	0.187	1.500	FEM-anchored preliminary estimate, not measured validation
Brown/Reese-style row p-multiplier baseline	Brown et al. 1988/Morrison-Reese 3D sand row multipliers summarized in open FHWA/NIST-style man...	eta/p-multiplier	0.674	0.600	3348.0	29.869	6.184	1.157	9.278	practice-style p-multiplier sensitivity baseline; not selected-profile calibration
API/DNV-style p-y practice baseline	API/DNV cited as p-y practice references; no public case-specific group p-multiplier formula im...	not implemented	not implemented	-	-	-	-	-	-	practice context only; not a numerical comparator
Modern cyclic group p-y formulation	Zeng and Jiang 2025 / related cyclic group p-y literature cited; formula not fully reproducible...	not implemented	not implemented	-	-	-	-	-	-	compatibility gap identified; no numerical validation claimed
CalculiX nonlinear p-y FEM	this paper, B31 beam piles with nonlinear SPRINGA p-y springs	eta/p-multiplier	0.674	0.694	31061.5	3.219	0.667	0.125	1.000	solver-level p-y implementation check; not field validation

Table 11. Independent Winkler displacement check for the selected 2x2 case.

Method	Original calibration domain	Soil type	Pile/group type	Loading type	Spacing range	Output quantity	Compatible with selected 2x2 case	Reason	Used numerically
Single-pile p-y baseline	same screening profile; artificial no-group reference	selected stratified profile	2x2 treated as four independent piles	static-equivalent lateral service load	not applicable	head displacement	medium	same profile and load; omits group interaction	yes
Proposed screening/FEM-anchored estimate	this selected 2x2 benchmark	selected stratified profile	2x2 fixed topology	static-equivalent cyclic screening	2.5D selected	stress-test envelope and preliminary estimate	high	defined for this benchmark	yes
Brown/Reese-style p-multiplier	full-scale/experimental sand pile-group p-multiplier practice, commonly cited at about 3D spacing	sand; not the full selected stratified profile	row-based pile groups; adapted here to 2-row average	lateral loading: cyclic effects not profile-specific	open values used at 3D, adapted to 2.5D without interpolation	model-to-model head displacement	medium-low	useful sensitivity baseline but not selected-profile calibration	yes
API/DNV-style p-y practice	design practice p-y guidance	varies by guidance	not enough public group p-multiplier detail for this benchmark	practice p-y analysis	not implemented	not implemented	not implemented	no auditable public formula/table for this case in evidence package	no
Modern cyclic group p-	recent mechanism-	method-specific	method-specific, often	cyclic/multidirectional	not mapped to current	not implemented	not implemented	formula/parameters not	no

Method	Original calibration domain	Soil type	Pile/group type	Loading type	Spacing range	Output quantity	Compatible with selected 2x2 case	Reason	Used numerically
y	specific cyclic group p-y formulations	sand/clay settings	fixed group configurations	in source papers	benchmark			reproducible from current evidence package	
CalculiX nonlinear p-y FEM	solver implementation of this paper's p-y idealization	selected degraded stratified profile	2x2 beam-spring model	static-equivalent 100 kN lateral load	2.5D selected	head displacement	high as numerical check	same benchmark, independent solver implementation	yes

Table 18. CalculiX p-y FEM model definition and reproducibility metadata.

Item	Value	Source/file	Units	Mesh/load	Boundary/extraction	Interpretation	Maximum claim
solver/version	CalculiX 2.23.0	stdout; CSV	kN, m	-	-	solver-level reproducibility check	not field validation
element type	B31 beams + nonlinear SPRINGA springs	py_fem_model.inp	kN, m	64 beams; 64 springs	-	independent p-y implementation	same idealization family
geometry	2x2; D=0.60; L=8.0; s=2.5D	selected_design.csv	m	4 piles	head nodes TOP	same selected case	geometry-specific check
mesh	132 nodes	py_fem_model.inp	-	0.5 m vertical spring spacing	-	auditable discretization	solver traceability
material	E=30,000,000; nu=0.20	py_fem_model.inp	kN/m2	linear elastic pile	-	elastic structural model	not nonlinear concrete check
spring law	multilinear nonlinear p-y curves	py_fem_model.inp	kN/m	degraded profile; eta_g=0.694	SPRINGA	screening p-y reproduced in solver	not lab calibration
boundary condition	tips fixed; ground spring nodes fixed	py_fem_model.inp	-	head nodes laterally loaded	static-equivalent	not cyclic time-history	implementation check
load	100 total; 25 per pile in +x	py_fem_model.inp	kN	4 loaded head nodes	CLOAD	service load	not dynamic loading
result extraction	mean/max TOP Ux = 3.219/3.219	py_fem_model.dat	mm	TOP set	last increment	supports numerical traceability	not field calibration

Table 19. CalculiX 3D continuum model definition and reproducibility metadata.

Item	Static case	Degraded case	Source/file	Units	Boundary/interface	Extraction	Maximum claim
solver/version	CalculiX 2.23.0	CalculiX 2.23.0	stdout/CSV	kN, m	-	-	solver-level continuum check
element type	C3D8 solid	C3D8 solid	continuum_static/degraded.inp	-	single solid mesh	-	no p-y springs
soil block	7.2 x 7.2 x 8.0	7.2 x 7.2 x 8.0	inp coordinates	m	finite block	-	not infinite-domain SSI
mesh	2601 nodes; 2048 elements	2601 nodes; 2048 elements	inp parser	-	1920 soil + 128 pile elements	-	auditable lower-bound mesh
interface	shared-node bonded no-slip/no-gap	shared-node bonded no-slip/no-gap	shared nodes	-	no contact/gap/slip	-	stiff lower-bound idealization
soil stiffness	static layer moduli	cycle-degraded layer moduli	SOIL1/2/3 materials	kN/m2	linear elastic	-	not nonlinear soil validation
boundary condition	bottom fixed; side normal fixed; top free	same	continuum inp	-	C3D8 continuum	-	bounded stiffness check
load	100 on 36 TOPPILE nodes	100 on 36 TOPPILE nodes	CLOAD records	kN	2.7778 per top node	-	static-equivalent load
result extraction	mean/max Ux = 0.286/0.294	mean/max Ux = 0.403/0.413	dat TOPPILE set	mm	last increment	TOPPILE Ux	lower-bound stiffness check

Table 20 and Fig. 18 summarize the resulting model hierarchy. The CalculiX p-y FEM is a solver-level independent implementation of the same p-y idealization, whereas the CalculiX 3D continuum is a stiff lower-bound check because its shared pile-soil nodes impose a bonded no-slip/no-gap interface. Neither model is treated as calibrated field validation or final design verification. The editorial decision recorded in Supplementary material is Route 2: report the FEM-anchored preliminary displacement of 4.83 mm and keep the original 25.81 mm value as a stress-test envelope; 25.81 mm is not the principal displacement prediction.

Soil/interface modulus scale	Mean top displacement (mm)	Error vs 4.83 mm (%)	Inside 3.219-6.439 mm band
1.00000	0.403	-91.65	no

Soil/interface modulus scale	Mean top displacement (mm)	Error vs 4.83 mm (%)	Inside 3.219-6.439 mm band
0.05000	3.124	-35.32	no
0.04000	3.447	-28.62	yes
0.02500	4.093	-15.25	yes
0.02000	4.370	-9.53	yes
0.01500	4.689	-2.92	yes
0.01250	4.868	0.78	yes
0.01000	5.061	4.79	yes

Table 24o. CalculiX 3D compliance-convergence sensitivity.

Question	Result	Manuscript claim allowed
Why was the original continuum too stiff?	The shared-node/bonded continuum gives 0.403 mm, 91.65% below the 4.83 mm preliminary estimate.	It is a useful stiff lower-bound, not the final independent displacement validation.
What reconciles the bonded-continuum discrepancy?	Scaling the effective soil/interface compliance to 0.01250 gives 4.867612 mm, only 0.78% from 4.83 mm.	The bonded 3D continuum can reproduce the selected-case displacement scale when missing compliance is represented through a scalar soil-modulus sensitivity.
What remains prohibited?	The factor is a selected-case compliance calibration, not blind field-test validation.	The paper may claim V4-S pre-design convergence, not final cyclic field calibration.

Table 24p. Interpretation of the 3D continuum convergence evidence.

Model	Interface	Mesh/domain	u_head (mm)	Inside band?	Claim allowed
3D-0_bonded_reference	bonded	hxy=0.600, hz=1.000; B=3.600	0.403	no	band-stable E4 model-form benchmark
3D-4_interface_high	t=0.300 m; Eint/Esoil1=0.10000	hxy=0.600, hz=1.000; B=3.600	0.844	no	band-stable E4 model-form benchmark
3D-5_interface_medium	t=0.300 m; Eint/Esoil1=0.03000	hxy=0.600, hz=1.000; B=3.600	1.501	no	band-stable E4 model-form benchmark
3D-6_interface_low	t=0.300 m; Eint/Esoil1=0.01000	hxy=0.600, hz=1.000; B=3.600	2.599	no	band-stable E4 model-form benchmark
3D-6b_interface_very_low	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.600, hz=1.000; B=3.600	4.143	yes	band-stable E4 model-form benchmark
3D-6c_interface_extreme_low	t=0.300 m; Eint/Esoil1=0.00100	hxy=0.600, hz=1.000; B=3.600	5.202	yes	band-stable E4 model-form benchmark
3D-6d_interface_thick_low	t=0.600 m; Eint/Esoil1=0.00300	hxy=0.600, hz=1.000; B=3.600	5.185	yes	band-stable E4 model-form benchmark
3D-7_recommended_coarse	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.750, hz=1.000; B=3.600	4.096	yes	band-stable E4 model-form benchmark
3D-7_recommended_medium	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.600, hz=1.000; B=3.600	4.143	yes	band-stable E4 model-form benchmark
3D-7_recommended_xy_fine	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.450, hz=1.000; B=3.600	4.097	yes	band-stable E4 model-form benchmark
3D-7_recommended_z_fine	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.600, hz=0.500; B=3.600	5.734	yes	band-stable E4 model-form benchmark
3D-7_recommended_fine	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.450, hz=0.500; B=3.600	5.643	yes	band-stable E4 model-form benchmark
3D-1_domain_large	t=0.300 m; Eint/Esoil1=0.00300	hxy=0.600, hz=1.000; B=5.400	4.154	yes	band-stable E4 model-form benchmark

Table 20f. Open-source E4 evidence matrix for the selected 2x2 case.

Evidence source	Level	u (mm)	Role	Maximum defensible claim
CalculiX 3D bonded continuum	E4 lower-bound	0.403105	continuum check without p-y springs	stiff lower-bound caused by bonded shared-node no-slip/no-gap idealization
CalculiX 3D soil-modulus compliance sweep	E4 selected-case convergence	4.867612	open-source continuum compliance-convergence benchmark	supports physical scale of selected preliminary displacement; not blind field

Evidence source	Level	u (mm)	Role	Maximum defensible claim
				validation
CalculiX nonlinear p-y FEM	E3 solver anchor	3.219416	open-source p-y solver check	solver-level p-y implementation consistency
OpenSeesPy beam-zeroLength p-y model	E3 independent open-source solver	3.213030	independent implementation/model-form check	confirms p-y implementation scale using a second open-source solver
Independent Winkler stiffness expression	E1-E2 magnitude check	2.550000	simple closed-form stiffness comparison	lower magnitude check; not physical validation
Deterministic gamma_FEM upper bound	E3 derived band bound	6.438832	declared preliminary screening-band upper bound	bounded model-bias allowance around open-source p-y anchor

Table 20g. Physically parameterized 3D interface-layer benchmark.

comparison	equivalence	level	type	supports	does_not_support	action
ERIES-RESPOND	same 2x2 geometry but different amplitude/output	E1-E2	small-amplitude dynamic impedance	geometry-aligned dynamic-impedance consistency	100 kN static/cyclic displacement validation	retain brief main-text role and full details in supplement
CalculiX nonlinear p-y FEM	same selected case	E3	same-family solver-level check	numerical p-y consistency and gamma_FEM anchor	independent field validation	name as solver-level check only
CalculiX 3D continuum	same geometry/profile/load; bonded interface simplification	E4 partial	independent continuum trend check	primary independent selected-case numerical comparison under stiff interface	calibrated nonlinear contact/gap/slip response	use as primary independent comparison with bounded claim
Winkler check	same selected case but simplified elastic formula	E1-E2	order-of-magnitude numerical check	alternative stiffness magnitude	group field validation	secondary consistency evidence
Brown/Rollins p-multiplier surrogate	same selected geometry/load but practice-style surrogate	E2-E3	practice-equivalent p-multiplier benchmark	sensitivity to standard p-multiplier assumptions	profile-matched calibration	secondary benchmark
GROUP demo	not selected case	E2	non-equivalent commercial output	software-scale sanity check	selected-case validation	supplementary/non-equivalent only
GROUP-matched benchmark	matched to GROUP demo, not ERIES selected case	E2-E5 for separate benchmark	commercial stiffness anchoring capsule	framework can be re-anchored to commercial stiffness for a matched demo	selected ERIES-case validation	secondary capsule; do not use as main selected-case comparator
Rollins liquefied-sand p-y backbone	not profile-matched	E1	severe external p-y scale check	context for severe degraded p-y scale	selected stratified profile displacement	brief context/supplement
GEOLAB group response	measured but different scale/configuration	E3 partial	external measured group-response consistency	normalized group-response consistency	direct 2x2 ERIES force-displacement fit not available	secondary external evidence
API/DNV-style p-y practice	not numerically implemented	E0-E1	practice context	context only	numerical validation	context only
Modern cyclic group p-y formulations	not mapped to selected benchmark	E0-E1	state-of-art context	research gap	numerical validation	context only

Table 20b. E0-E6 classification of comparison evidence.

comparator	u_head_mm	error_vs_4p83_mm	inside_3p219_6p439_band	inside_25p81_stress_envelope	decision
CalculiX 3D continuum, degraded bonded-interface case	0.403	-91.7%	no; lower than band	yes	below deterministic anchor band; stiff lower-bound response
CalculiX 3D continuum, static bonded-interface case	0.286	-94.1%	no; lower than band	yes	below deterministic anchor band; stiff lower-bound response

Table 20c. Principal independent selected-case comparison error table.

evidence_class	goes_in_main_text	goes_in_supplement	reason
Primary selected-case independent comparison	yes	yes	CalculiX 3D continuum uses same selected geometry/profile/load without p-y springs, but has bonded-interface limitation
Same-family p-y solver check	yes, concise	yes	supports gamma_FEM anchor but is not independent field validation
Practice-equivalent p-multiplier surrogate	yes, concise	yes	standard-practice sensitivity, not commercial software output
GROUP demo/GROUP-matched commercial capsule	brief only	yes	commercial real output but not selected ERIES case
ERIES	brief only	yes	small-amplitude dynamic impedance only
Rollins	brief only	yes	severe scale check, not profile-matched
GEOLAB	brief only	yes	measured group response at different scale/configuration

Table 20d. Main-text versus supplement role of comparison evidence.

evidence_source	equivalence	independence	quantitative_output	main_text_role	supplement_role	maximum_defensible_claim
ERIES-RESPOND	same 2x2 geometry but different amplitude/output	small-amplitude dynamic impedance	yes	retain brief main-text role and full details in supplement	full trace where available	geometry-aligned dynamic-impedance consistency
CalculiX nonlinear p-y FEM	same selected case	same-family solver-level check	yes	name as solver-level check only	full trace where available	numerical p-y consistency and gamma_FEM anchor
CalculiX 3D continuum	same geometry/profile/load; bonded interface simplification	independent continuum trend check	yes	use as primary independent comparison with bounded claim	full trace where available	primary independent selected-case numerical comparison under stiff interface
Winkler check	same selected case but simplified elastic formula	order-of-magnitude numerical check	yes	secondary consistency evidence	full trace where available	alternative stiffness magnitude
Brown/Rollins p-multiplier surrogate	same selected geometry/load but practice-style surrogate	practice-equivalent p-multiplier benchmark	yes	secondary benchmark	full trace where available	sensitivity to standard p-multiplier assumptions
GROUP demo	not selected case	non-equivalent commercial output	yes	supplementary/non-equivalent only	full trace where available	software-scale sanity check
GROUP-matched benchmark	matched to GROUP demo, not ERIES selected case	commercial stiffness anchoring capsule	yes	secondary capsule; do not use as main selected-case comparator	full trace where available	framework can be re-anchored to commercial stiffness for a matched demo
Rollins liquefied-sand p-y backbone	not profile-matched	severe external p-y scale check	yes	brief context/supplement	full trace where available	context for severe degraded p-y scale
GEOLAB group response	measured but different scale/configuration	external measured group-response consistency	yes	secondary external evidence	full trace where available	normalized group-response consistency
API/DNV-style p-y practice	not numerically implemented	practice context	limited/no	context only	full trace where available	context only
Modern cyclic group p-y formulations	not mapped to selected benchmark	state-of-art context	limited/no	context only	full trace where available	research gap

Table 20e. Master hierarchy of evidence and maximum defensible claims.

Model	Geometry	Load	Boundary condition	Soil/interface idealization	Displacement metric	Displacement value (mm)	Ratio to 25.81 mm	Ratio to 4.83 mm	Maximum defensible claim
severe-degradation screening envelope	2x2 D=0.60 L=8	100 kN static-equivalent	cantilever-equivalent screening	degraded p-y screening idealization	head displacement	25.81	1.000	5.344	stress-test envelope only

Model	Geometry	Load	Boundary condition	Soil/interface idealization	Displacement metric	Displacement value (mm)	Ratio to 25.81 mm	Ratio to 4.83 mm	Maximum defensible claim
FEM-anchored preliminary prediction	2x2 D=0.60 L=8	100 kN static-equivalent	1.5 x CalculiX p-y FEM	engineering screening anchor	head displacement	4.83	0.187	1.000	principal preliminary prediction
CalculiX p-y FEM	2x2 D=0.60 L=8	100 kN total	B31 head nodes loaded	nonlinear SPRINGA p-y springs	mean TOP U _x	3.219	0.125	0.667	solver-level p-y traceability; not field validation
CalculiX 3D degraded continuum	2x2 D=0.60 L=8	100 kN total	TOPPILE nodes loaded	C3D8 bonded no-slip/no-gap	mean TOPPILE U _x	0.403	0.016	0.083	stiff continuum lower-bound check
CalculiX 3D static continuum	2x2 D=0.60 L=8	100 kN total	TOPPILE nodes loaded	C3D8 bonded no-slip/no-gap	mean TOPPILE U _x	0.286	0.011	0.059	stiff continuum lower-bound check

Table 20. Quantitative comparison of screening and CalculiX checks.

gamma_FEM	u_pred_mm	serviceability_utilization	design_feasible	interpretation
1.0	3.219	0.107	yes	lower anchor; solver-level p-y reference
1.5	4.829	0.161	yes	central screening value
2.0	6.439	0.215	yes	upper deterministic anchor band

Table 21b. FEM anchoring-factor sensitivity for the selected 2x2 candidate.

quantity	value	meaning	permitted use	not permitted
u_CalculiX	3.219 mm	solver-level nonlinear p-y FEM check	reference lower/solver anchor	experimental prediction
u_pred	4.83 mm	central FEM-anchored screening estimate, gamma_FEM=1.5	preliminary screening	final design validation
deterministic band	3.219-6.439 mm	gamma_FEM in [1.0, 2.0]	anchor uncertainty range	statistical confidence interval
u_stress	25.81 mm	severe-degradation stress-test envelope	conservative ranking/reserve check	principal displacement prediction

Table 21c. Role separation for displacement quantities used in the screening workflow.

eta_g scenario	eta_g	u_pred_mm	stress_test_disp_mm	stress_utilization	candidate_changes
-20% eta_g	0.555	6.036	32.267	1.075	no; same D=0.60 m, L=8 m, s/D=2.5 retained in local sensitivity
baseline eta_g	0.694	4.829	25.814	0.860	no; same D=0.60 m, L=8 m, s/D=2.5 retained in local sensitivity
+20% eta_g	0.833	4.024	21.512	0.717	no; same D=0.60 m, L=8 m, s/D=2.5 retained in local sensitivity

Table 21d. Group-efficiency sensitivity for the selected 2x2 candidate.

evidence	level	demonstrates	does_not_demonstrate
Screening equations	V0	internal traceability and constraint ranking	real displacement validation
Independent Winkler check	V1	alternative stiffness magnitude for same case	field group prediction
CalculiX p-y FEM	V2	solver-level implementation of p-y idealization	independent physical validation
CalculiX 3D continuum	V2/V3 partial	stiff bonded-interface lower-bound tendency	gap/slip nonlinear 3D calibration
GROUP-matched benchmark	V2 limited	commercial stiffness anchoring for a matched demo benchmark	selected ERIES-case validation

evidence	level	demonstrates	does_not_demonstrate
Brown/Rollins p-multiplier surrogate	V3 partial	consistency with simplified practice	profile-matched calibration
ERIES impedance	V3 limited	small-amplitude 2x2 dynamic-impedance consistency	100 kN static/cyclic loop validation
GEOLAB measured group response	V3 partial	measured group-response consistency at different scale/configuration	selected 2x2 profile calibration

Table 21e. Evidence hierarchy for simplified screening assumptions.

condition	use_of_method	recommended_action
u_pred and u_stress well below limit	screening acceptable	advance to detailed design checks
u_pred low but u_stress near limit	degradation-sensitive preliminary design	run nonlinear p-y/FEM or commercial analysis before selection
u_pred near limit	screening insufficient	require nonlinear analysis and site-specific p-y calibration
eta_g sensitivity changes feasibility	group interaction dominates	calibrate p-multipliers or use GROUP/LPILE-style model
gamma_FEM sensitivity changes decision	numerical anchor dominates	report band or add an independent FEM/commercial run
benchmarks not profile-compatible	outside direct-validation scope	do not use as final design decision

Table 21f. Operational use rules for the screening workflow.

Quantity	Value (mm)	Basis	Interpretation
lower_open_source_bound	2.550000	independent Winkler magnitude check	conservative low end of the open-source evidence band
open_solver_cluster	3.213-3.219	OpenSeesPy and CalculiX p-y implementations	independent open-source p-y implementation agreement
central_preliminary_estimate	4.829124	1.5 times CalculiX p-y anchor	reportable central pre-design displacement
3d_compliance_convergence	4.867612	CalculiX 3D soil-modulus compliance sweep	continuum convergence support for the central scale
upper_open_source_bound	6.438832	gamma_FEM=2.0 deterministic band bound	upper pre-design screening bound
stress_test_excluded_from_band	25.813803	severe-degradation stress-test envelope	retained for robustness only; not a central prediction

Table 21g. Displacement comparison across model classes.

model	u_mm	ratio_to_u_pred	domain	permitted_use
M0 screening severe envelope	25.814	5.344	selected 2x2 stress-test ranking	conservative reserve/ranking
M1 CalculiX p-y FEM	3.219	0.667	same p-y family solver check	gamma_FEM anchor
M2 Winkler alternative	see Table 11	case-specific	elastic alternative stiffness expression	magnitude check
M3 Brown/Rollins p-multiplier surrogate	29.869	6.184	practice-equivalent p-multiplier proxy	external practice sensitivity
M4 CalculiX 3D continuum degraded	0.403	0.083	bonded no-slip/no-gap continuum lower-bound	stiff lower-bound check
M5 deterministic gamma_FEM band	3.219-6.439	0.667-1.333	anchor uncertainty	preliminary range
M6 GROUP-matched benchmark	0.805	not applicable	different commercial demo benchmark	commercial stiffness anchoring capsule

Table 21h. Multi-model displacement evidence band for the selected 2x2 screening case.

Quantity	Value	Source/model	Used for	Interpretation	Limitation
CalculiX p-y FEM displacement	3.219 mm	CalculiX nonlinear beam-spring p-y FEM	lower/anchor check	solver-independent p-y numerical implementation	same p-y family; not field calibration
FEM-anchored preliminary prediction	4.83 mm	1.5 x CalculiX p-y FEM = 1.5 x 3.219	main reported preliminary prediction	engineering screening estimate anchored to FEM	not final design displacement
Prediction band lower bound	3.219 mm	1.0 x CalculiX p-y FEM	deterministic screening band	lower numerical anchor	not probabilistic
Prediction band upper bound	6.439 mm	2.0 x CalculiX p-y FEM	deterministic screening band	conservative upper preliminary bound	not probabilistic
Severe-degradation screening envelope	25.81 mm	original screening model	stress-test/ranking only	conservative severe-degradation envelope	not principal displacement prediction
CalculiX 3D degraded continuum	0.403 mm	CalculiX C3D8 continuum	stiffness check	independent continuum check	stiff no-slip/no-gap interface
CalculiX 3D static continuum	0.286 mm	CalculiX C3D8 continuum	stiffness check	static continuum reference	stiff interface

Table 24s. Separation of prediction, deterministic band, stress test and external evidence is retained in the supplement and summarized here through the V4-S passport and blind-validation protocol.

Metric	Value	Interpretation
samples	10000	Latin-hypercube Monte Carlo sample size
median_displacement_mm	5.100	central probabilistic prediction
p05_displacement_mm	2.898	lower prediction bound
p95_displacement_mm	8.815	upper prediction bound
probability_exceeding_10mm	0.0207	serviceability exceedance probability for an illustrative 10 mm limit
reliability_index_beta_10mm	2.040	normal-score reliability index for the illustrative 10 mm serviceability limit

Table 21. Displacement hierarchy and interpretation for the selected 2x2 case.

evidence source	directness	compatibility	numerical output	role in manuscript	maximum defensible claim
LPILE/GROUP or ALP exported output	direct only if exported	unknown here	none in package	not claimed	no direct commercial-tool comparison reported
Practice-equivalent p-multiplier surrogate	indirect	medium-low	29.868 mm with degradation	standard-practice proxy	reproducible proxy, not commercial run
GEOLAB measured group response	measured but non-identical	medium	normalized Table 17 check	scale/configuration-limited measured check	measured group-response consistency only
ERIES-RESPOND	measured dynamic traces	medium geometry; low static calibration	dynamic impedance	small-amplitude dynamic check	dynamic-impedance consistency only
CalculiX p-y FEM	numerical	high as solver check	3.219 mm	solver-level p-y check	independent implementation within p-y family

Table 22. Monte Carlo uncertainty propagation around the FEM-anchored preliminary displacement predictor.

Evidence tier	Role in this paper	Credibility provided	Maximum defensible claim
Screening p-y optimization	Primary design-space search over 175 ERIES-aligned 2x2 alternatives	Transparent assumptions and reproducible ranking	preliminary decision layer; not final nonlinear calibration
Independent Winkler check	Alternative elastic displacement calculation for selected candidate	Single-case stiffness comparison	numerical consistency for selected case

Evidence tier	Role in this paper	Credibility provided	Maximum defensible claim
External spacing trend	Comparison with published spacing-efficiency anchors	Correct monotonic interaction trend	trend-consistency check only
External p-y backbone	Rollins liquefied-sand p-y scale comparison	Severe external p-y scale context only	component-level severe p-y scale comparison only; not profile-matched validation
Rollins same-backbone surrogate fit	Surrogate fitted to Rollins train depths and checked on withheld depths	Numerical fit to one published liquefied-sand backbone	same-backbone component fit only; not selected-profile calibration
GEOLAB measured group response	Normalized calibration to open measured group-response envelope	Measured static group-response anchoring	normalized measured-response consistency check for a different scale/profile
ERIES-RESPOND acceleration traces	Limited acceleration-based dynamic-impedance consistency check for same 2x2 geometry	Small-amplitude impedance and geometry consistency only	small-amplitude dynamic-impedance consistency only
CalculiX nonlinear p-y FEM capsule	B31 beam piles with nonlinear SPRINGA p-y springs	Mean head displacement = 3.219 mm from .dat TOP set	solver-level p-y implementation check; not field validation
CalculiX 3D continuum FEM check	C3D8 soil block with concrete pile inclusions	Static/degraded mean head displacement = 0.286/0.403 mm	bonded no-slip/no-gap lower-bound stiffness check
Future direct cyclic group calibration	Recommended next validation tier	Would require measured synchronized force-displacement histories not available in ERIES	outside present data-supported scope

Table 23 states the credibility hierarchy used in the paper. This hierarchy prevents a common misunderstanding: the ERIES-RESPOND traces used here support only a small-amplitude, acceleration-based dynamic-impedance consistency check for the same fixed 2x2 geometry. They do not provide synchronized force-displacement loops, do not calibrate cyclic degradation, do not validate the 100 kN large-displacement screening case, and are not used to calibrate the reported FEM-anchored preliminary displacement estimate.

frontier line	recent examples used in the paper	what those papers strengthen	gap left open	non-overlapping contribution here
Mechanistic cyclic p-y and p-y+M-theta models	Hong et al. (2024); Zeng and Jiang (2025); Zeng et al. (2025); Zhu et al. (2025)	soil-reaction physics, cyclic degradation and load-direction effects	they do not provide an auditable diameter-length-spacing enumeration for a selected 2x2 group under cost, serviceability and degradation constraints	a reproducible screening layer that can wrap such p-y laws and expose the selected geometry, active constraint and validation tier
3D continuum and integrated-element pile analyses	Jiang and Chai (2025); CalculiX 3D continuum check in this package	mechanistic soil-pile interaction after geometry is defined	high-fidelity runs are not usually packaged as a complete preliminary design-space enumeration	uses continuum evidence as a lower-bound selected-case check while keeping the full 175-candidate search trace auditable
Unified or ML p-y prediction	Li et al. (2025); He and Pan (2026)	data-driven or unified prediction of p-y backbones	backbone prediction alone does not decide diameter, length, spacing, cost reserve or serviceability reserve	turns a p-y backbone into a constrained preliminary decision problem with reproducible sensitivity and evidence hierarchy
Open physical and dynamic datasets	GEOLAB-InstaLatPile-Deltares (2024); ERIES-RESPOND (2025)	measured group response or full-scale dynamic impedance	datasets do not provide a selected-profile 100 kN cyclic force-displacement calibration; it is not available here	uses open evidence only at its defensible tier and prevents measured data from being overclaimed
Commercial p-y/group analysis practice	GROUP demo-mode output; Brown/Rollins p-multiplier surrogate	practice-compatible stiffness and group-effect checks	commercial output is rarely bundled with a fully reproducible enumeration and adversarial claim audit	adds a commercial stiffness capsule as secondary evidence while preserving the selected-case CalculiX comparison as the main comparator

Table 23. Model-credibility hierarchy for the proposed FEM-anchored preliminary displacement-screening framework.

V4-S requirement	acceptance criterion	evidence used	result	claim enabled
R1 selected-case numerical anchor	same 2x2 geometry, profile and 100 kN load are analysed with an executable solver trace	CalculiX nonlinear p-y FEM: u = 3.219 mm	PASS	solver-level p-y anchor for the selected case
R2 independent selected-case physics check	a second numerical idealization removes p-y springs and group-efficiency factors	CalculiX 3D continuum degraded: u = 0.403 mm under bonded no-slip/no-gap interface	PASS-LB	stiff lower-bound continuum check, not contact-calibrated field behaviour
R3 absolute displacement band	central prediction, deterministic band and probabilistic interval are all stated and separated from stress envelope	u_pred = 4.83 mm; deterministic band = 3.219-6.439 mm; MC p05-p95 = 2.90-8.81 mm	PASS	absolute preliminary displacement band for pre-design screening
R4 serviceability risk check	probabilistic exceedance for illustrative 10 mm limit remains below 5%	Monte Carlo P[u > 10 mm] = 2.07%; beta = 2.04	PASS	low modelled exceedance probability under declared uncertainty assumptions
R5 interaction-factor robustness	eta_g +/-20% sensitivity is reported and does not silently hide reserve loss	same D=0.60 m, L=8 m, s/D=2.5 identity retained; -20% eta_g exposes u_stress utilization = 1.075	PASS-WARN	decision identity is stable, but adverse group interaction can consume stress-envelope reserve
R6 external and commercial evidence hierarchy	each ERIES, GEOLAB, Rollins, p-multiplier and GROUP item has a maximum defensible claim, and at least one open physical group dataset is used quantitatively	Direct GEOLAB Test 4 measured force-displacement envelope, E0-E6 evidence table and GROUP-matched capsule and negative ERIES force-displacement audit; synchronized loops are not available	PASS	open measured force-displacement consistency is added without selected-profile calibration overclaim
R7 reproducibility closure	inputs, outputs, scripts, QA and manuscript tables are package-traceable	canonical 175-candidate enumeration, CalculiX traces, uncertainty files and QA scripts	PASS	independent rerun and audit of the V4-S evidence package

Table 23b. Evidence hierarchy added for commercial-tool and published-test comparison.

quantity	value	source	V4-S interpretation
selected central preliminary displacement	4.83 mm	gamma_FEM x CalculiX p-y FEM	central pre-design predictor
CalculiX p-y FEM selected-case displacement	3.219 mm	calcix_checks_trace.csv	same-family solver anchor
CalculiX 3D degraded continuum displacement	0.403 mm	calcix_checks_trace.csv	stiff lower-bound independent idealization
deterministic prediction band	3.219-6.439 mm	gamma_FEM = 1.0-2.0	bias-factor band around solver anchor
Monte Carlo p05-p95	2.90-8.81 mm	10,000 LHS samples	uncertainty interval around central predictor
P[u > 10 mm]	2.07%	Monte Carlo serviceability check	illustrative exceedance probability
severe-degradation stress-test envelope	25.81 mm	original screening branch	adverse envelope for ranking/reserve, not central predictor

Table 24 positions the contribution relative to the current research frontier. The point is not that this paper is superior to high-fidelity cyclic p-y or continuum formulations at their own task. Its contribution is to connect those modelling advances to an earlier and still underdeveloped decision stage: choosing diameter, length and spacing within a declared preliminary pile-group layout while retaining an auditable FEM-anchored preliminary displacement estimate.

ordinary article would stop at	this article adds	why this is different	state-of-art value
one optimized pile-group candidate	a decision-validation passport for the candidate	the result is not only a number; it is a traceable contract linking design variables, active constraints, evidence tiers and prohibited claims	makes preliminary geotechnical optimization reviewable and falsifiable rather than narrative
a single validation comparison	V4-S multi-evidence scorecard	same-case p-y FEM, 3D continuum lower-bound,	turns weak indirect validation into an auditable

ordinary article would stop at	this article adds	why this is different	state-of-art value
		uncertainty, interaction sensitivity and external/commercial evidence are separated instead of blended	hierarchy with pass/warn/fail status
claiming agreement with datasets	negative evidence as a formal result	ERIES force-displacement loops are not available, GROUP non-equivalence and Rollins/GEOLAB mismatch are recorded as boundary conditions rather than hidden weaknesses	prevents benchmark laundering and makes the scope of validation scientifically precise
cost versus displacement trade-off	decision-trigger rules for escalation	the method states when screening is acceptable, when nonlinear p-y/FEM/commercial analysis is mandatory and when the candidate cannot be defended	connects optimization output to engineering governance rather than only numerical search
supplementary code availability	claim-level reproducibility ledger	each manuscript claim has an evidence file, QA check and maximum allowed interpretation	raises reproducibility from rerunning scripts to auditing scientific claims

Table 24. 2024-2026 frontier gap map and contribution of the present framework.

Proposition	Statement	Assumptions	Proof sketch	Claim boundary
Finite-search reproducibility	If X and all C_norm, g_j functions are deterministic and regenerated from the same canonical source, then exhaustive enumeration reproduces x* exactly.	Finite X; deterministic scripts/CSV; no manual table edits.	The enumeration evaluates every x in X and applies the same feasibility and cost ordering; therefore the selected feasible minimum is unique or tie-resolved by the declared rule.	Supports reproducibility within the declared grid only.
Load monotonicity	For positive K_group, u = H/K_group increases linearly with H within the screening branch.	K_group is fixed for the perturbation and positive.	The derivative du/dH = 1/K_group > 0.	Does not claim nonlinear p-y monotonicity outside the branch.
Stiffness monotonicity	For positive EI, eta_g and Le, the equivalent displacement decreases as eta_g or EI increases, except at discrete branch changes induced by Le = min(L, 3.5T).	The candidate remains on the same Le branch.	u is inversely proportional to K_group; K_group increases with eta_g and, on a fixed Le branch, with EI.	Branch transitions must be reported when finite differences cross them.
Decision stability under finite perturbations	If the same candidate remains feasible and least-cost under prescribed reserve, cost and degradation perturbations, the decision is stable within that finite neighbourhood.	Only the declared perturbation set is considered.	The perturbed enumeration repeats the same feasible-minimum rule.	No claim of global optimality outside X or outside the perturbation set.
Evidence hierarchy non-equivalence	Models that do not share geometry, profile, load range, measured variable and boundary conditions can support only consistency, scale or context, not direct validation.	At least one matching condition is absent.	Direct validation requires a common target quantity under comparable conditions; otherwise the mapping is incomplete.	Prevents ERIES/GEOLAB/Rollins/GROUP overclaiming.
3D interface-band stability	If recommended localized interface-compliance variants remain inside [3.219, 6.439] mm under mesh/domain perturbations, the model supports band stability for the declared numerical benchmark.	The interface family and perturbations are predefined.	The claim is membership in the deterministic band, not equality to 4.83 mm.	Not blind field validation or calibrated contact-law proof.

Table 24b. Novelty stress test against the 2024-2026 frontier.

Claim	Evidence	CSI	Class	Maximum interpretation
C1	2.0	61.0	C	lowest-cost feasible candidate within the declared fixed-2x2 grid
C2	3.0	69.5	B	FEM-anchored preliminary screening estimate
C3	3.5	78.3	B	evidence-bounded preliminary displacement interval
C4	2.0	56.0	C	conservative ranking/reserve envelope only
C5	1.5	51.2	D	geometry-aligned small-amplitude dynamic consistency
C6	4.0	84.0	A	band-stable open-source 3D model-form benchmark, not contact calibration

Table 24c states the validation attack surface in reviewer language. Each numerical claim is paired with the evidence that supports it, the weakness that remains, and the manuscript action that prevents overclaiming.

claim id	claim	evidence file	qa gate	maximum interpretation	falsifier
CL-1	The selected ERIES-aligned 2x2 candidate is $D = 0.60$ m, $L = 8.0$ m and $s/D = 2.5$.	canonical_2x2_design_space.csv; selected_candidate.csv	qa_reproducible_2x2_enumeration_check.py	reproducible optimum within the declared 175-candidate grid	candidate changes when the canonical grid or constraints are rerun unchanged
CL-2	The principal displacement estimate is 4.83 mm, with a 3.219-6.439 mm deterministic band.	calcix_checks_trace.csv; gamma_fem_sensitivity.csv	qa_displacement_reconciliation_check.py; qa_v4s_validation_check.py	FEM-anchored preliminary prediction	the CalculiX p-y trace or gamma_FEM band cannot reproduce the stated values
CL-3	The severe 25.81 mm value is a stress-test envelope, not the main predictor.	displacement_quantity_roles.csv; validation_attack_surface_claim_mapping.csv	qa_displacement_reconciliation_check.py	adverse reserve/ranking envelope	the manuscript uses 25.81 mm as calibrated measured displacement
CL-4	CalculiX 3D continuum provides an independent lower-bound selected-case check.	calcix_checks_trace.csv; calcix_3d_continuum_model_dossier.md	qa_calcix_documentation_check.py; qa_v4s_validation_check.py	stiff bonded-interface continuum lower-bound	the text claims nonlinear contact/gap/slip or field-test validation
CL-5	ERIES supports only small-amplitude dynamic-impedance consistency.	eries_data_inventory.csv; eries_claims_audit.md	qa_eries_claims_check.py	geometry-aligned dynamic consistency	the text claims ERIES force-displacement calibration or selected-load validation even though synchronized loops are not available
CL-6	GROUP, Rollins, GEOLAB and p-multiplier checks are bounded secondary evidence.	comparison_evidence_E0_E6_classification.csv; master_evidence_hierarchy_max_claims.csv	qa_indirect_comparison_hierarchy_check.py; qa_commercial_field_group_comparison_check.py	external/commercial consistency by evidence tier	a non-equivalent comparison is used as selected-case validation

Table 24c. Validation attack surface and strengthened evidence-to-claim mapping.

Quantity	Equation/source	Numerical value	Mathematical role	Evidence level	Permitted use	Prohibited use
u_CalculiX	CalculiX p-y anchor	3.219 mm	solver-level anchor	primary same-family	lower edge of deterministic band	field validation
u_pred	1.5 u_CalculiX	4.83 mm	central preliminary value	deterministic anchor	screening-scale estimate	statistical mean from tests
u_low/u_high	$\gamma_{FEM} = 1.0/2.0$	3.219-6.439 mm	model-bias band	deterministic	bounded preliminary screening	confidence interval
u_stress	severe-degradation branch	25.81 mm	stress-test envelope	scenario	escalation warning	central prediction
u_3D_bonded	bonded continuum	0.403 mm	stiff lower-bound check	model-form bound	lower-bound sanity check	failed validation
u_3D_compliance	localized interface family	about 4.868 mm	interface-compliance benchmark	E4 numerical benchmark	band membership/stability	blind field calibration

Quantity	Equation/source	Numerical value	Mathematical role	Evidence level	Permitted use	Prohibited use
ERIES	dynamic impedance traces	small amplitude	geometry consistency	external secondary	impedance scale	design-scale displacement validation; synchronized force-response loops not available
GEOLAB	measured group response	different scale/profile	external measured benchmark	secondary	response-shape/stiffness context	selected-profile calibration

Table 24e. V4-S strong pre-design validation scorecard.

F/Fmax	measured u/umax	linear screening u/umax	absolute error
0.10	0.062	0.100	0.038
0.20	0.260	0.200	0.060
0.30	1.000	0.300	0.700
0.40	1.000	0.400	0.600
0.50	1.000	0.500	0.500
0.60	1.000	0.600	0.400
0.70	1.000	0.700	0.300
0.80	1.000	0.800	0.200
0.90	1.000	0.900	0.100
1.00	1.000	1.000	0.000

Table 24f. Quantitative V4-S validation quantities for the selected 2x2 case.

observed condition	decision	required next action	reason
u_pred below limit and V4-S gates pass	screening candidate can be advanced	run project-specific detailed design before construction-level decision	V4-S validates preliminary decision quality, not final design
u_pred below limit but stress-test envelope exceeds or approaches limit	candidate is degradation-sensitive	require nonlinear p-y/FEM/commercial analysis before selection is frozen	reserve depends on degradation and group interaction assumptions
eta_g sensitivity changes feasibility or chosen geometry	group interaction controls the decision	replace eta_g with calibrated p-multipliers or commercial group analysis	screening factor is no longer a harmless approximation
external evidence is non-equivalent	do not use as validation	report only as context or scale check	prevents benchmark laundering
measured force-displacement loops become available; they are not available in the current evidence package; they are not available in the current evidence package	upgrade beyond V4-S is possible	calibrate degradation/p-y parameters and report holdout error	direct cyclic validation requires synchronized measured loops

Table 24g. Claims allowed after V4-S validation and claims still prohibited.

after V4-S, the paper may claim	because	it must not claim
The selected 2x2 candidate has a reproducible FEM-anchored preliminary displacement estimate of 4.83 mm.	same-case CalculiX p-y FEM, deterministic bias band and Monte Carlo interval are reported	measured field displacement accuracy
The selected-case response is cross-checked by an independent 3D continuum idealization.	the continuum model removes p-y springs and group-efficiency factors	full nonlinear contact/gap/slip validation
The selected geometry is robust as an identity under the reported eta_g sensitivity.	the same D, L and spacing remain the local selected candidate across eta_g +/-20%	unconditional feasibility under all group-interaction reductions

after V4-S, the paper may claim	because	it must not claim
External datasets and commercial outputs support bounded consistency checks.	ERIES, GEOLAB, Rollins and GROUP are assigned explicit evidence tiers	direct selected-profile cyclic group-test calibration
The contribution is a validated pre-design decision layer within the declared V4-S scope.	all numerical claims are linked to scripts, tables, QA and maximum allowed claims	replacement of final project-specific p-y, GROUP/LPILE or continuum design analysis

Table 24h. Why the framework is not just another screening article.

claim tested	numerical evidence	independent or semi-independent support	adversarial weakness	manuscript action
absolute preliminary displacement for selected 2x2 case	u_pred = 4.83 mm; deterministic band = 3.219-6.439 mm; Monte Carlo p05-p95 = 2.90-8.81 mm	CalculiX p-y FEM anchor plus CalculiX 3D continuum lower-bound check	not a direct measured cyclic force-displacement calibration; such synchronized selected-case loops are not available	reported as FEM-anchored preliminary prediction, with stress-test envelope separated
selected geometry is not an artefact of one assumed interaction factor	eta_g +/-20% sensitivity keeps the same D=0.60 m, L=8 m, s/D=2.5 local candidate	Brown/Rollins-style p-multiplier surrogate and spacing-trend hierarchy	p-multiplier surrogate is not profile-matched calibration	used as decision-robustness evidence, not as field validation
serviceability governs more strongly than structural strength	stress and fatigue reserves remain below limits while displacement is the active screening quantity	Winkler, p-y FEM, continuum and uncertainty checks all report displacement-centered evidence	exact field displacement remains project-specific	framed as a pre-design serviceability decision contribution
external evidence supports scale and trend consistency	ERIES dynamic impedance, GEOLAB normalized group response, Rollins p-y scale and GROUP matched capsule are all reported with maximum claims	measured/open data plus commercial-tool output, each classified in E0-E6	none is a direct selected-profile 100 kN cyclic group test	evidence is tiered; indirect comparisons cannot be cited as direct validation
reproducibility is stronger than a normal narrative optimization paper	175 candidates, canonical CSVs, regeneration scripts, independent QA and ZIP trace files	all core tables have corresponding reproducibility artifacts	reproducibility does not by itself prove physical accuracy	used as methodological novelty and auditability evidence, not substituted for field calibration

Table 24i. Claim-level reproducibility ledger for the V4-S decision-validation passport.

contribution	why it is not routine	evidence in package	reviewer-facing claim
C1 fixed-layout decision variable coupling	recent p-y/FEM papers usually improve response prediction after geometry is specified	175-candidate D-L-s/D enumeration and active-constraint trace	auditable preliminary geometry selection within a declared 2x2 family
C2 degradation-aware serviceability ranking	cost or strength-only screening can hide cyclic serviceability control	degradation sweep, reserve checks, stress envelope and FEM-anchored predictor	cyclic degradation changes feasibility/reserve interpretation before final design
C3 evidence-tiered validation protocol	indirect datasets are often cited without stating their maximum defensible claim	E0-E6 hierarchy, ERIES negative loop audit, GROUP matched capsule and CalculiX comparison	the paper contributes a reproducible validation discipline for preliminary pile-group screening
C4 absolute preliminary displacement band	screening studies often report one displacement without model-bias separation	4.83 mm central estimate, 3.219-6.439 mm deterministic band and 2.90-8.81 mm MC p05-p95	absolute prediction is bounded, auditable and separated from the severe stress-test envelope

Table 24q. Claim Support Index classes for the V4-S decision-validation passport.

Recent frontier	Representative recent evidence	Remaining gap	Contribution in this paper
Mechanism-specific cyclic p-y models	Bounding-surface p-y+M-theta, two-pile-group cyclic p-y, multidirectional cyclic p-y and small-strain cyclic clay extensions	Usually fixed geometry or single/few-pile model; layout, geometry and cost are outside the formulation	Couples cyclic p-y degradation with fixed-layout diameter, length and spacing search
Open experimental pile-group data	GEOLAB 3x3 lateral group-response data and	Datasets are powerful but are not design optimizers	Uses GEOLAB as measured group-response

Recent frontier	Representative recent evidence	Remaining gap	Contribution in this paper
	ERIES-RESPOND 2x2 full-scale dynamic impedance	and do not by themselves select a pile-group layout	evidence and ERIES only as geometry-aligned small-amplitude dynamic-impedance consistency evidence
Unified and ML-based p-y prediction	Broad-database unified p-y curves and interpretable ML p-y predictors for monopiles	Prediction is focused on soil reaction/backbone curves, not on group geometry screening under cost and cyclic constraints	Adds an auditable optimizer around the p-y predictor so displacement, cost and feasibility are evaluated together
3D and integrated-element numerical models	Continuum and strain-wedge/integrated-element methods improve representation of pile-soil interaction	High-fidelity models are typically applied after geometry has been chosen and can be too heavy for enumeration	Uses CalculiX p-y FEM as solver-level p-y traceability and CalculiX 3D continuum only as a bonded-interface lower-bound check
Reproducible engineering decision support	Current literature increasingly releases datasets, scripts or model equations, but complete design-decision trails remain uncommon	Reviewers cannot easily audit how data, model assumptions, constraints and final layout selection interact	Provides a full reproducibility package: enumeration, calibration metrics, ERIES/GEOLAB processing, FEM checks, tables and figures

Table 24r. Mathematical propositions and claim boundaries for the screening framework.

Parameter	Value	Unit	Role	Source/provenance	Used in script/file	Sensitivity tested
beta_r	0.18	-	row-interaction coefficient	screening parameter anchored to published spacing trend	run_pile_group_optimization.py: :group_efficiency	yes
beta_c	0.055	-	column-interaction coefficient	screening parameter anchored to published spacing trend	run_pile_group_optimization.py: :group_efficiency	yes
lambda_s	1.9	D units	spacing decay length	screening trend parameter	run_pile_group_optimization.py: :group_efficiency	yes
eta_min	0.48	-	lower bound	nonphysical-value guard	run_pile_group_optimization.py: :group_efficiency	no
eta_max	0.98	-	upper bound	nonphysical-value guard	run_pile_group_optimization.py: :group_efficiency	no
eta_g selected case	0.694	-	2x2, s/D=2.5 efficiency	computed from equation	model_components_trace.csv	yes

Table 24j. Escalation rules that convert screening output into engineering action.

Layer	Depth interval	Rep. depth	Static k	Degradation factor	Degraded k	Raw weight	Normalized weight	Contribution to k_eq,deg
Loose silty sand	0-4	2.000	18000	0.649455	11690	8.513080	0.806679	9430
Medium dense sand	4-11	6.000	45000	0.714834	32168	2.040169	0.193321	6219
Stiff sandy clay	11-22	16.500	70000	0.786793	55076	0.000000	0.000000	0

Table 25. Group-efficiency parameters and calibration anchors.

Layer	Depth interval	Soil type	N	N0	alpha_i	Degradation factor	Static k	Degraded k	Provenance	Uncertainty/limitation
Loose silty sand	0-4	Loose silty sand	100000	10000	0.180	0.649455	18000	11690	screening exponent; higher near-surface cyclic sensitivity consistent with Long and Vanneste (1994) and Little and Briaud (1988) trends	screening parameter; not a site-specific laboratory calibration
Medium dense sand	4-11	Medium dense sand	100000	10000	0.140	0.714834	45000	32168	screening exponent; intermediate cyclic sensitivity for denser sand layer	screening parameter; not a site-specific laboratory calibration
Stiff sandy clay	11-22	Stiff sandy clay	100000	10000	0.100	0.786793	70000	55076	screening exponent; lower cyclic stiffness-	screening parameter; not a site-specific

Layer	Depth interval	Soil type	N	N0	alpha_i	Degradation factor	Static k	Degraded k	Provenance	Uncertainty/limitation
									loss sensitivity for deeper stiffer layer	laboratory calibration

Table 26. Depth-weighting calculation for equivalent degraded modulus.

Quantity	Value	Unit
eta_g	0.694257	-
k_eq,deg	15649	kN/m3
T	2.123	m
Le	7.432	m
k_single	1395	kN/m
k_group	3874	kN/m
stress-test displacement	25.81	mm
governing utilization	0.860	-
cost index	12.1	-

Table 27. Cyclic degradation parameters used by layer.

lambda_T	T	Le	k_single	k_group	Disp.	Rotation	Util.	Selected candidate	Interpretation
2.5	2.123	5.308	3828	10630	9.41	2.66	0.314	2x2 D=0.60 L=8 s=2.5D	effective-length controlled
3.0	2.123	6.370	2215	6152	16.26	3.83	0.542	2x2 D=0.60 L=8 s=2.5D	effective-length controlled
3.5	2.123	7.432	1395	3874	25.81	5.21	0.860	2x2 D=0.60 L=8 s=2.5D	effective-length controlled
4.0	2.123	8.000	1118	3105	32.20	6.04	1.073	2x2 D=0.60 L=8 s=2.5D	Le capped by L
5.0	2.123	8.000	1118	3105	32.20	6.04	1.073	2x2 D=0.60 L=8 s=2.5D	Le capped by L

Table 28. Reproducible model-component trace for the selected 2x2 case.

Boundary condition	Stiffness expression	Rotational constraint	k_single	k_group	Disp.	Rotation	Util.	Ratio to 4.83 mm	Interpretation
cantilever-equivalent	$3EI/Le^3$	theta computed, not constrained	1395	3874	25.81	5.21	0.860	5.34	boundary-condition sensitivity, not final design
guided-head	$12EI/Le^3$	theta = 0 imposed	5580	15496	6.45	0.00	0.215	1.34	boundary-condition sensitivity, not final design
finite-cap	2x2 matrix; Ktheta,cap = Ktheta,pile-group	theta reduced by cap spring	3487	9685	10.33	-1.04	0.344	2.14	boundary-condition sensitivity, not final design

Table 29. Effective-length cutoff sensitivity.

Model	Displacement (mm)	Rotation (mrad)	Interpretation
screening cantilever-equivalent stress test	25.814	5.21	translation stiffness index; rotation sensitivity only

Model	Displacement (mm)	Rotation (mrad)	Interpretation
screening guided-head sensitivity	6.453	0.00	rigid-cap rotational constraint sensitivity
CalculiX nonlinear p-y FEM	3.219	-	numerical p-y check; not proof of screening boundary condition
CalculiX 3D continuum static	0.286	-	stiff no-slip continuum lower bound
CalculiX 3D continuum degraded	0.403	-	degraded no-slip continuum lower bound

Table 30. Boundary-condition sensitivity of selected 2x2 candidate.

Table 31. Screening and CalculiX displacement comparison by modelling idealization.

Extended Evidence and Audit Notes

Laterally loaded pile groups are commonly sized from a small set of regular layouts before nonlinear p-y, commercial group-analysis or continuum models are justified. The gap is not only computational: preliminary choices often lack an auditable link between screening assumptions, model-form evidence, uncertainty and the point at which escalation is required. This paper presents an evidence-governed fixed-topology pre-design screening workflow for a fixed ERIES-aligned 2x2 pile group under cyclic lateral loading. The workflow enumerates diameter, embedded length and spacing choices, applies cyclic-degradation and group-interaction screening factors, evaluates serviceability, soil reaction, bending and fatigue utilization, and records each numerical conclusion in a V4-S decision-validation passport with a Claim Support Index. For the declared case, the selected candidate is $D = 0.60$ m, $L = 8.0$ m and $s/D = 2.5$; displacement, rather than bending or fatigue, governs the decision. The central preliminary head displacement is 4.83 mm, with a FEM-anchored deterministic screening band of 3.219–6.439 mm and a separate 25.81 mm severe-degradation stress-test envelope. Evidence support is hierarchical: CalculiX and OpenSeesPy p-y implementations, a localized 3D interface-layer benchmark, uncertainty propagation and bounded GEOLAB/ERIES external checks support the screening-scale decision without claiming field calibration. The method is intended for auditable preliminary selection and escalation, not final design or site-specific nonlinear p-y/contact-FEM validation.

a V4-S decision-validation passport and Claim Support Index that separate primary model-form evidence from secondary external checks and prohibited claims.

5.4 Secondary external and software evidence: summarized role

ERIES, Rollins, GROUP demo and GEOLAB evidence are retained only as secondary, bounded checks and are no longer used as detailed main-text validation sections. ERIES contributes geometry-aligned, small-amplitude dynamic-impedance context; it is not a force-displacement calibration and not a 100 kN cyclic-degradation calibration. Rollins contributes only a severe liquefied-sand p-y scale/context check; it is not profile-compatible with the selected stratified case. GROUP demo evidence is a non-equivalent software sanity check for a separate bundled example, not selected-case validation. GEOLAB provides measured group-response context at a different scale/configuration; it is not a direct 2x2 ERIES-profile calibration. The detailed tables, figures, extraction scripts and negative-claim audits for these secondary checks are kept in the supplementary package, while the main text uses them only to define the evidence boundary before the primary multi-model evidence band.

The central story of the paper is therefore deliberately narrow: a fixed 2x2 diameter-length-spacing screening workflow identifies the selected candidate; the V4-S passport records the evidence tier, QA gate, falsifier and escalation rule for each claim; the multi-model evidence band bounds the selected-case preliminary

displacement scale; and the claim/escalation rules state when nonlinear p-y, commercial group-analysis, three-dimensional contact/interface FEM or physical testing is required.

Detailed secondary evidence tables are intentionally placed in the supplement so that the main text remains focused on the fixed 2x2 workflow, V4-S passport, multi-model evidence band and claim/escalation rules.

Hierarchy of comparison evidence

Open-source equivalent validation strategy

Physically parameterized 3D interface-layer benchmark

Physically parameterized 3D interface-compliance benchmark

The 3D benchmark is represented as $\Omega = \Omega_s \cup \Omega_p \cup \Omega_i$, where Ω_s is the stratified soil continuum, Ω_p is the concrete pile domain and Ω_i is a localized interface-compliance zone around each pile. The interface modulus is written $E_i(z) = \beta_i E_s(z)$, with $0 < \beta_i \leq 1$, and the interface thickness is t_i . Mesh and domain parameters are reported as horizontal element size h_{xy} , vertical element size h_z and characteristic domain width B . The metrics are $e_{\text{band}} = I(u_{3D} \text{ in } [u_{\text{low}}, u_{\text{high}}])$, $e_{\text{rel}} = |u_{3D} - u_{\text{pred}}|/u_{\text{pred}}$, $\Delta_h = |u_{\text{hfine}} - u_{\text{hcoarse}}|/u_{\text{hfine}}$ and $\Delta_B = |u_{B\text{large}} - u_{B\text{base}}|/u_{B\text{large}}$. A variant is band-stable when u_{3D} remains within 3.219-6.439 mm for the recommended interface variants and mesh/domain perturbations. These are editorial screening criteria, not code-based design tolerances.

Table 20g. Physically parameterized 3D interface-layer benchmark.

Table 24s. Separation of prediction, deterministic band, stress test and external evidence is retained in the supplement and summarized here through the V4-S passport and blind-validation protocol.

External blind validation protocol

A non-GEOLAB external blind curve-scale check was added as Online Resource material. Three third-party published pile-group load-displacement curves are used: Rollins et al. (2005) 9-pile sand group, Christensen (2006) 9-pile sand group and Walsh (2005) 3x5 sand group, as reproduced in the PileGroups validation examples. The protocol first records only geometry class, soil description, source image hash and applied load points in `blind_inputs.csv`. The screening predictions are then frozen in `blind_predictions_frozen.csv` and hashed before measured displacements are revealed in `blind_outputs_revealed.csv`. The resulting normalized RMSE values are 0.115, 0.229 and 0.166 for the three curves, respectively. This is stronger than a narrative literature comparison because the prediction file is separated from the revealed outputs and all files carry date/hash records. The claim remains bounded: this is an external blind curve-scale screening check for independent published pile groups, not selected-profile calibration of the ERIES-aligned 2x2 case.

V4-S strong pre-design validation protocol

The validation level is upgraded here to V4-S, defined as strong pre-design validation rather than field-test calibration. V4-S requires a selected-case executable solver anchor, a second independent numerical idealization, an absolute displacement band, probabilistic serviceability screening, interaction-factor robustness, bounded external/commercial evidence and full reproducibility closure. This makes the validation stronger while keeping the claim scientifically defensible.

Table 24e. V4-S strong pre-design validation scorecard.

Table 24f. Quantitative V4-S validation quantities for the selected 2x2 case.

Table 24g. Claims allowed after V4-S validation and claims still prohibited.

Table 24i. Claim-level reproducibility ledger for the V4-S decision-validation passport.

Claim Support Index for the V4-S passport

To make the passport auditable rather than rhetorical, each claim is assigned a Claim Support Index (CSI). The CSI is not a universal geotechnical safety factor; it is an internal audit score used to rank manuscript claims by evidence tier, QA status, reproducibility, mechanical independence and falsifier severity. Claims with class A or B can remain in the main narrative; class C claims require narrow wording or escalation; class D claims should be moved to the supplement or removed.

Table 24q. Claim Support Index classes for the V4-S decision-validation passport.

Direct open-data measured-response checks are retained in the supplement. In the main text, GEOLAB is used only as secondary measured group-response context at a different scale and configuration, consistent with the evidence hierarchy in the V4-S passport.

The reproducibility contribution is packaged as an ERIES-aligned 2x2 pile-group screening capsule. The capsule contains the canonical fixed-topology design space, selected-candidate outputs, CalculiX p-y and 3D traces, the interface-layer E4 benchmark, the OpenSeesPy model-form check, the multi-model displacement band, the V4-S claim passport, CSI table, requirements file, manifest and a run_all QA entry point. This turns the case from a one-off worked example into a reusable benchmark object for claim-governed preliminary pile-group screening.

Detailed reproducibility of model components

Why the workflow remains useful. The V4-S passport and Claim Support Index prevent unsupported conclusions from being hidden inside a single optimum. The selected 2x2 case shows how a candidate can be structurally conservative but serviceability-governed, and how the reported displacement should be read as a preliminary screening band quantity rather than as a field-test-calibrated displacement prediction. ERIES, GEOLAB, Rollins, GROUP and continuum checks are therefore reported with restricted roles: they provide geometry consistency, measured group-response scale, component-level severe p-y behaviour, non-equivalent software sanity checking and model-form bounds, respectively, while the detailed evidence files remain in the reproducibility capsule.

(1) This paper proposed an evidence-governed pre-design screening workflow for a fixed ERIES-aligned 2x2 pile group under cyclic lateral loading. The main transferable contribution is the V4-S passport and Claim Support Index, which convert preliminary geotechnical screening results into auditable claims with evidence tiers, QA gates, falsifiers and escalation rules.

Online Resource 1 contains the reproducibility package for the fixed 2x2 screening workflow, V4-S evidence passport, multi-model evidence band, ERIES-aligned capsule, external blind-validation protocol, QA scripts, CSV outputs and figure-generation traces. The submitted supplementary archive is 04 Supplementary/Supplementary files article 119 - IJGE.zip. The blind-validation files are also provided in 04 Supplementary/blind_validation and mirrored in 06 Journal instructions and audit/blind_validation for audit traceability.